BSE Mining Module — MSc thesis

Do Chilean copper mines respond to ENSO forecasts?

The forecast signal, the impact season, and the absence of an anticipatory response

Chapter draft · 8 June 2026 · internal working document

ENSO is, in principle, the part of climate risk that a mine could see coming. The El Niño–Southern Oscillation is genuinely predictable months ahead, whereas an isolated storm is not; so if ENSO-relatedness matters for adaptation, the most direct channel is this predictability — a reliable seasonal forecast should let an exposed operation act before the shock arrives. This chapter tests whether that channel operates. It does not. The ENSO state carries a majority-probability forecast signal months ahead, but on its own the index predicts any single mine's output only weakly, and where the signal was both available and actionable, mines show no anticipatory response — neither in production nor in the permitting record. Adaptation in the sector is real and large, but it runs on a decade-scale capital clock, decoupled from any single forecast. The predictability that should make ENSO special buys no anticipatory advantage in practice.

1. The forecast signal: available for the season that matters

Each month the CPC/IRI consensus assigns a probability to each ENSO phase for every upcoming three-month season. The object a mine should act on is not the season in which the index first crosses ±0.5°C (the *onset*), but the forecast for the season in which its operations are actually disrupted: the austral winter rainy season (JJA) for flood-exposed mines in central Chile, and the austral summer (DJF) for drought-exposed mines in the north.

Read against that impact season, a majority-probability forecast signal was present for **every one of the five episodes**. The El Niño flood season (JJA) first crossed 50% two to three months ahead — at 55–75%, peaking 93–97%; the La Niña drought season (DJF) first crossed 50% three to six months ahead — at 51–65%, peaking 84–98%. What was *not* reliably forecast was the precise onset: three of the five onsets (2010, 2020, 2023) never crossed 50% before they arrived. The contrast is the whole of Section 1.

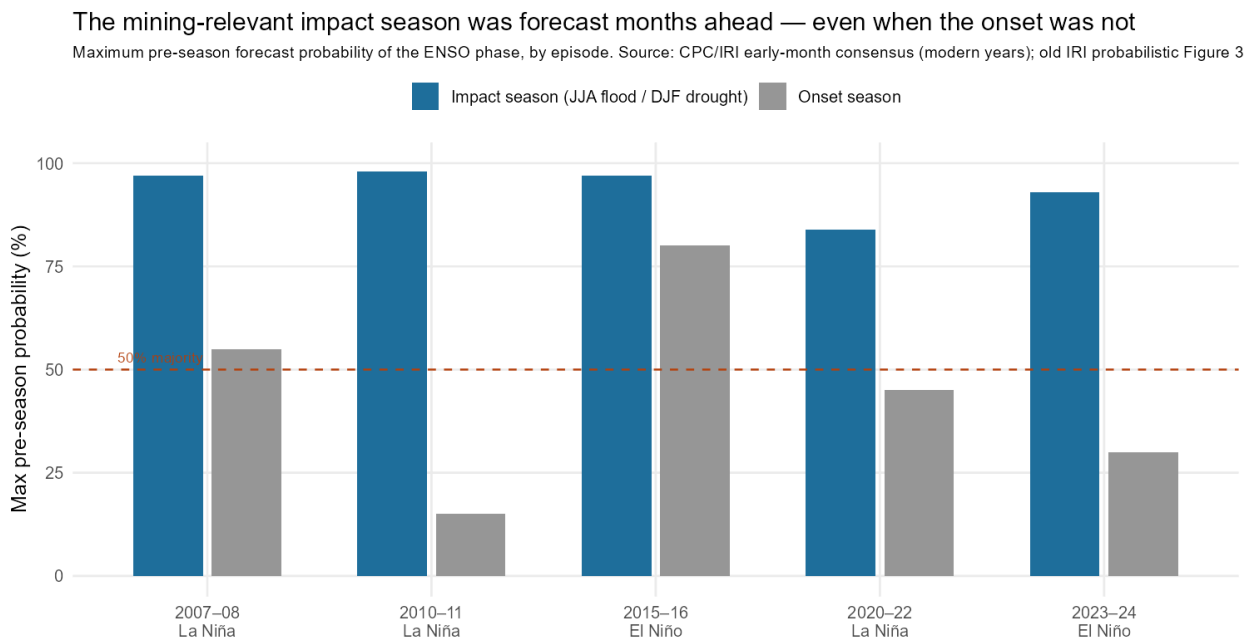


Figure 1. Maximum pre-season forecast probability of the ENSO phase, onset season vs mining-impact season, for each episode. Impact-season probability (blue) clears the 50% line for all five; onset probability (grey) does not for 2010, 2020, 2023. Source: CPC/IRI early-month consensus tables (modern years); old IRI probabilistic Figure 3 tables (2007, 2010).

The 2023 case is the cleanest illustration. The AMJ 2023 onset — the season the regression treats as the event — was never forecast above 50%. But P(El Niño) for JJA 2023, the season that threatens the rainy-season operations, stood at 75% in April 2023 and 93% by June. Pegging “forecastability” to the onset records a zero-month window; pegging it to the impact season records a confident, multi-month signal. The mine cares about the second.

Episode	Onset — first > 50%	Impact season — first > 50%
2007–08 La Niña	May 2007 (1 mo, 55%)	DJF: Sep 2007 (3 mo, 65%)
2010–11 La Niña	never	DJF: Jun 2010 (6 mo, 62%)
2015–16 El Niño	Mar 2014 (6 mo, 51%)	JJA 2015: Mar 2015 (3 mo, 55%)
2020–22 La Niña	never	DJF: Jul 2020 (5 mo, 51%)
2023–24 El Niño	never	JJA 2023: Apr 2023 (2 mo, 75%)

Table 1. First issuance month at which the phase probability crossed 50%, with lead time and the probability at that first crossing (Figure 1 plots the maximum pre-season probability — the two are complementary). Three onsets never qualify; all five impact seasons do.

Two qualifications complete the picture. First, the onset difficulty is genuine and physical — the spring predictability barrier makes fast or marginal crossings (the 2010 El Niño-to-La Niña flip; the marginal 2020 and 2023 onsets) hard to time. This is the defensible core of the “scarcity” observation, and it is confined to onset timing: even the La Niña drought impact seasons were forecast at 51–65% months ahead, so the disruption itself was foreseeable. Second, and more important for the thesis, even an accurate ENSO-state forecast is a poor predictor of a specific mine’s output: spring ONI does not predict subsequent winter flood-mine output (r = +0.10, p =

0.28, n = 92 mine-years). This is not a claim that ENSO is harmless: the companion report finds that when a flood or drought shock actually crosses an operational threshold, output falls materially — a wet regional month by about 4.7%, and persistent ENSO-aligned drought lowering annual output. The point is narrower — the ONI index is a noisy guide to whether such a shock will bind at a given mine, so an accurate forecast of the index is still only a weak forecast of that mine's output.

Two event definitions appear in this chapter by design. Section 1 asks about the onset crossing — the object the companion regression treats as the event — because the foreseeability of precisely that object is what is in question; the production tests (Section 2) classify ENSO years by the season that anticipates the impact — JJA ONI for El Niño, OND ONI for La Niña (La Niña established entering the austral-summer DJF dry season, so the September–November adaptation window precedes the classifier).

2. Output across the forecast windows

Section 1 fixed, for each episode, the window in which the forecast was live — the months ahead of the impact season when the majority-probability call was already on the table. A mine that acted on it would do the same thing in both channels: once the forecast arrived it would pull production forward and build stockpile, so output would step up from the quiet months before the window into the window itself, ahead of the loss. The signature is therefore not a high level during the window — a mine can run high all year for reasons unrelated to the forecast — but a rise from just before the window into it, beyond the step those same months show in an ordinary year. For each episode we compare output in the forecast window with the months immediately before it:

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step = mean dev(window months) − mean dev(pre-window months)
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where dev(i,m,y) = output(i,m,y) − norm(i,m) and norm(i,m) is the mine's mean log output in calendar month m across neutral-ONI years. Because norm enters both terms it cancels, so the step nets out the mine's level and any slow trend — an expansion, or El Abra's 2020–22 wind-down — while the neutral-year norm removes the ordinary seasonal change between the two periods. The flood window is April–May, before the JJA rainy season (pre-window: January–March); the drought window is September–November, before the DJF dry season (pre-window: June–August). As a yardstick, the same step in neutral years averages 0.0% for flood mines (s.d. 4.5) and −0.3% for drought mines (s.d. 5.4), so a genuine forecast response would have to clear roughly ±5%.

Event	Forecast window	Pre-window → window step
El Niño 2015–16 (flood)	Apr–May; ~69% by April	+5.7% (+9.5 → +15.1)
El Niño 2023–24 (flood)	Apr–May; 75% by April, 93% by June	−3.3% (−22.9 → −26.2)
La Niña 2007 (drought)	Sep–Nov; DJF signal 3–6 mo ahead	+2.1% (−0.3 → +1.8)
La Niña 2010 (drought)	Sep–Nov; rapid onset, short lead	−3.1% (+2.6 → −0.5)
La Niña 2022 (drought)	Sep–Nov; DJF signal 3–6 mo ahead	−0.2% (−0.2 → −0.4)

Table 2. The pre-window-to-window output step for each episode (window-months mean deviation minus pre-window mean), with deviations from the neutral-ONI-year mean for the same mine and calendar month, so level, trend, and ordinary seasonality net out. Benchmark: the step averages 0.0% (flood, s.d. 4.5) and −0.3% (drought, s.d. 5.4) in neutral years. Flood mines: Andina, Teniente, Anglo Sur, Los Pelambres; drought mines: Escondida, Collahuasi, Codelco Chuqui/RT/MH, Spence, El Abra.

No episode clears that bar. In the flood channel, 2015 — the episode with the earliest, clearest signal — stepped up +5.7% from January–March into April–May; it is the largest step in the set and the only one that leans positive, but at about 1.3 s.d. it sits only just above the flood band, and 2023 steps the other way (−3.3%), drifting down into the window rather than up. One modest, isolated lean in the best-forecast episode, contradicted by the other El Niño, is not the systematic build a forecast response would produce. The drought channel is settled by the step: the September–November windows ran above baseline only because these mines ran high all year, not because output rose into the window — the steps are +2.1% (2007), −3.1% (2010) and −0.2% (2022), all within ordinary year-to-year movement. The 2022 group step is itself an average of large offsetting mine-specific moves — Spence +16% as it ramped up, El Abra −9% as it wound down — exactly the trend noise the step is built to remove. In no episode does output rise into the forecast window beyond what the same months do in a normal year.

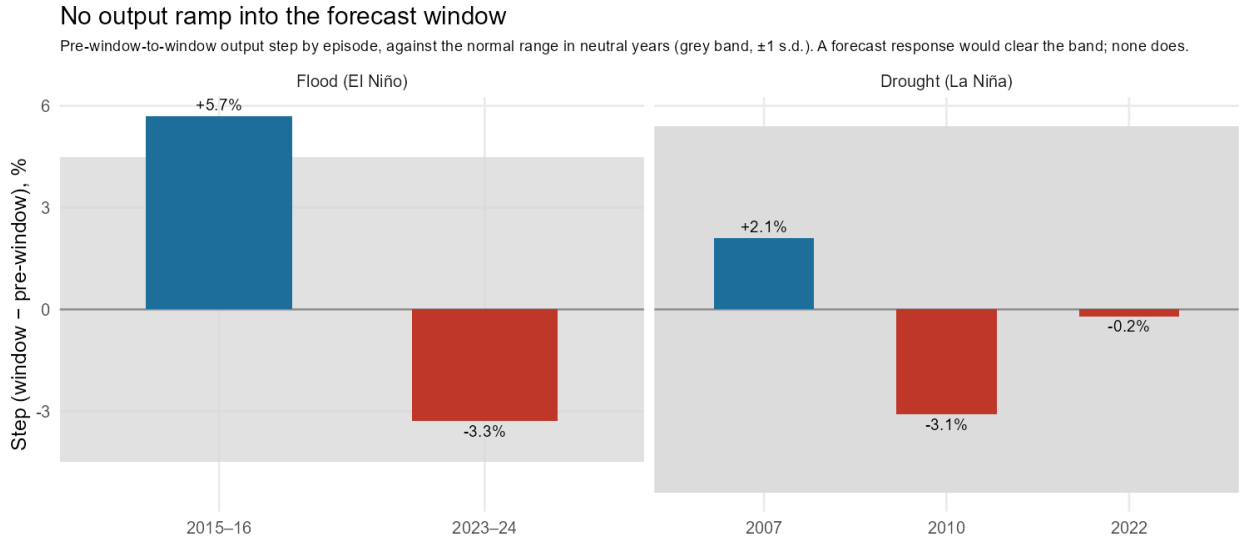


Figure 2. Pre-window-to-window output step, by episode, against the normal range in neutral years (grey band, ±1 s.d.). A forecast response would push an episode's step clearly above its band; none does — 2015 is the only positive lean and barely clears it, while 2023 steps down.

3. No response in the decision record

A production test only captures what reaches monthly tonnage. The complementary test asks whether mines made discrete, documented decisions — permit filings, capital commitments — in response to the forecast. We assembled the e-SEIA environmental-filing record and assigned each filing to an operation by región and proponent (titular) rather than by project name, which avoids the toponym collisions a name search invites. El Tesoro and Esperanza share a proponent and collapse to one Minera Centinela, and the Codelco Antofagasta divisions are separated by comuna so the drought-exposed Chuqui/RT/MH stays distinct from Gaby — giving 18 distinct operations. The record covers 16 of these, including all nine climate-exposed mines

(four flood, five drought); only Candelaria and Cerro Colorado fall outside it — neither is among the nine exposed mines used in the forecast-response tests (both do appear in the broader 19-mine output ledger). This is a sampling choice, not a statement about water exposure: Cerro Colorado is, if anything, the sample's most water-stressed operation (see Section 4).

Within that record, exactly two filings fall in the April–May windows of the two El Niño episodes — neither from a flood-exposed mine, and the only water-related one a US$0 administrative modification to a desalination project already under construction.

Mine	Date	Type	US$MM	Water	Project
Anglo American Norte	2015-04-20	DIA	90	no	Tailings disposal (Mantos Blancos)
Escondida	2015-04-06	DIA	0	admin mod	Desal supply — modification of existing project

Table 3. Every filing inside the two El Niño windows, across the 16 operations covered (all nine climate-exposed mines included). Neither is a flood-mine water or flood response.

The same test applied to the three La Niña episodes returns the same result. Across the September–November forecast windows — the months in which the drought signal was already live, ahead of the austral-summer dry season — none of the five drought-exposed mines filed a water-supply or drought-adaptation permit. Every filing they did lodge inside a window is routine and operational:

Mine	Date	Type	US$MM	Water	Project
Collahuasi	2007-11-30	DIA	14	no	Leaching-area adjustment
Codelco Chuqui	2010-09-23	DIA	0.2	no	Hospital wastewater treatment unit
Collahuasi	2010-10-19	DIA	30	no	Molybdenum plant upgrade
Collahuasi	2010-11-19	DIA	50	no	Rosario Sur ore extraction
Escondida	2020-08-13	DIA	0	no	Tailings-closure modification (withdrawn)

Table 4. Every filing by a drought-exposed mine inside the three La Niña forecast windows: Sep–Nov 2007, Jun–Nov 2010, Jul–Nov 2020 (window start = the month the impact-season forecast first exceeded 50%, Table 1). None is a water or drought-response filing; Spence and El Abra filed nothing in any window. (This lead-time window dates the 2020–22 episode to its 2020 crossing; Section 2's output test classifies the same episode by its 2022 impact season.)

None is a water or drought-response filing: Collahuasi's are a leaching-area adjustment and an ore-body extension, Chuquicamata's a hospital wastewater unit, Escondida's a withdrawn tailings-closure modification. The only genuine desalination filing falling inside any La Niña window is Candelaria's July 2010 EIA (US$270M) — an operation outside the 16-mine sample, and one the structural account already reads as predating the episodes. As in the El Niño windows, the drought decision-record holds no anticipatory response.

This is not a gap in the record; it is the finding — and it is close to mechanical, because permit and capital decisions run on multi-year cycles and will be absent from any three-to-five-month window. The informative pattern is in the full record: the sector's water-transition filings cluster

after the 2022 Water Code reform (El Abra, US$7.5bn, 2026; Zaldívar, US$1.2bn, 2023; Los Bronces, US$1.1bn, 2024; Gabriela Mistral, US$868M, 2024), while the pioneer — Escondida's first desalination plant, 2008 — preceded the 2015 El Niño by seven years. The clock is structural and regulatory, not the seasonal forecast.

4. Why: the structural clock

The mismatch is temporal. The forecast signal, where it exists, is good for weeks to months; the adaptation tool it would trigger — desalination capacity — takes years to permit, finance, and build, and then operates for decades. A mine cannot accelerate a multi-billion-dollar water pipeline in response to a two-month warning.

The sector is nonetheless adapting, structurally. Seawater rose from about 7% of mining water supply in 2012 to 36% in 2023, and COCHILCO projects roughly 68% by 2034. This decade-scale shift is keyed to the structural megadrought and the 2022 Water Code, not to any single ENSO event.

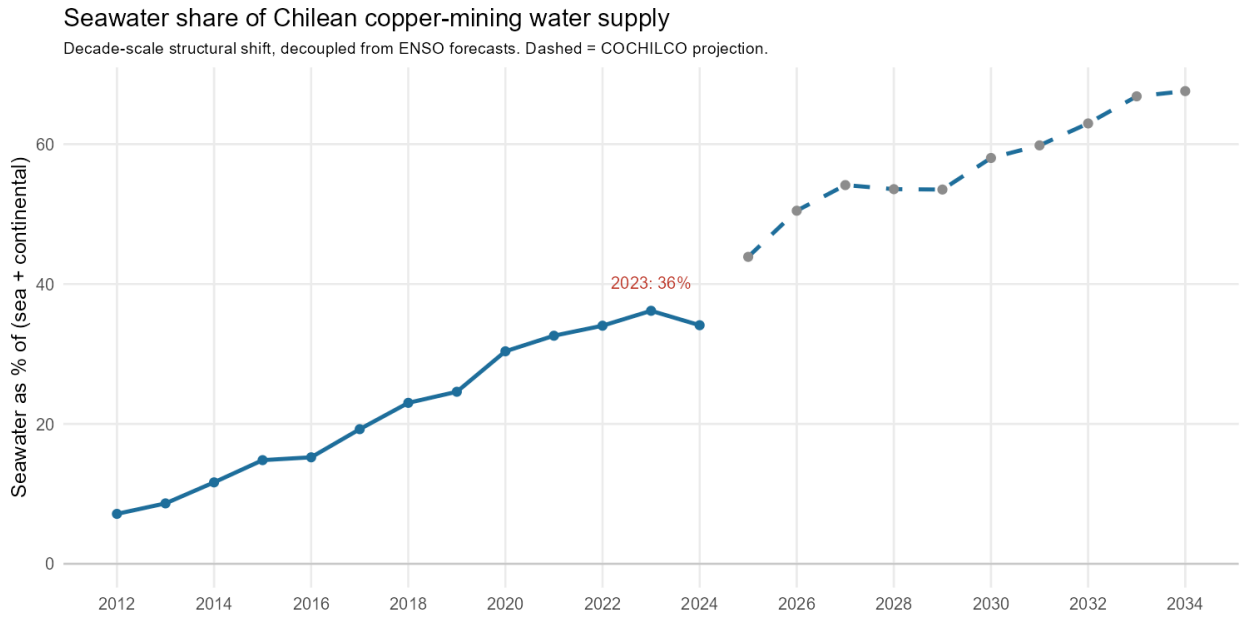


Figure 3. Seawater share of copper-mining water supply, 2012–2024 (actual) and 2025–2034 (COCHILCO projection).

The Los Pelambres paradox makes the decoupling concrete: the mine's desalination plant — built to hedge long-run drought — had its marine construction works disrupted by the 2023–24 El Niño swells, delaying commissioning by about a year. The adaptation tool was set back by the very kind of event it might have hedged, because it was designed for a different risk (drought) on a different clock (decadal).

Cerro Colorado (Tarapacá) is that clock in its limiting form. Structural water scarcity — a depleted aquifer and the non-renewal of its groundwater-extraction permit — forced the mine into closure in 2023; no seasonal forecast could have averted it. The northern mines that substituted to seawater in time keep operating; Cerro Colorado, still dependent on a contested

aquifer, did not. This is the distinction the chapter rests on: a seasonal ENSO shock disrupts output for a period and the mine recovers, whereas the slow, structural water constraint can be existential — and it binds on a multi-year clock that no forecast lead can shorten.

5. The regulatory record: water-scarcity decrees

Sections 2 and 3 looked for an anticipatory response inside the firm — in output and in its own permit filings. A third register sits outside the firm: the State's own water-scarcity declarations. Under the Water Code the Dirección General de Aguas (DGA) issues **decretos de zona de escasez** — dated, region-tagged, legally binding declarations that trigger water-allocation restrictions. If the regulator anticipated ENSO drought, these decrees would lead the deficit and track the forecast; if it merely reacts, they would lag it. We align the full DGA decree registry (2008–2026) to the July-centred (JJA) ENSO state and to the physical drought channel of the master ledger.

The direct link from the ENSO index to decree activity is weak: detrended against the steep secular rise of the instrument, national decrees correlate only $r \approx -0.22$ with JJA ONI (not significant on eighteen water years). But that weakness is a property of the policy layer, not of the underlying hydrology. Decomposed, the signal runs in a clear chain — JJA ONI grips physical drought stress ($r \approx -0.46$), and physical drought stress drives decrees ($r \approx +0.45$), each link moderate and, on the region-year panel, highly significant (Table 5). The ENSO signal is real in the water balance; it is attenuated, lagged, and made sticky as it passes through the administrative process. Decrees confirm ENSO drought once it has bound — they do not forecast it.

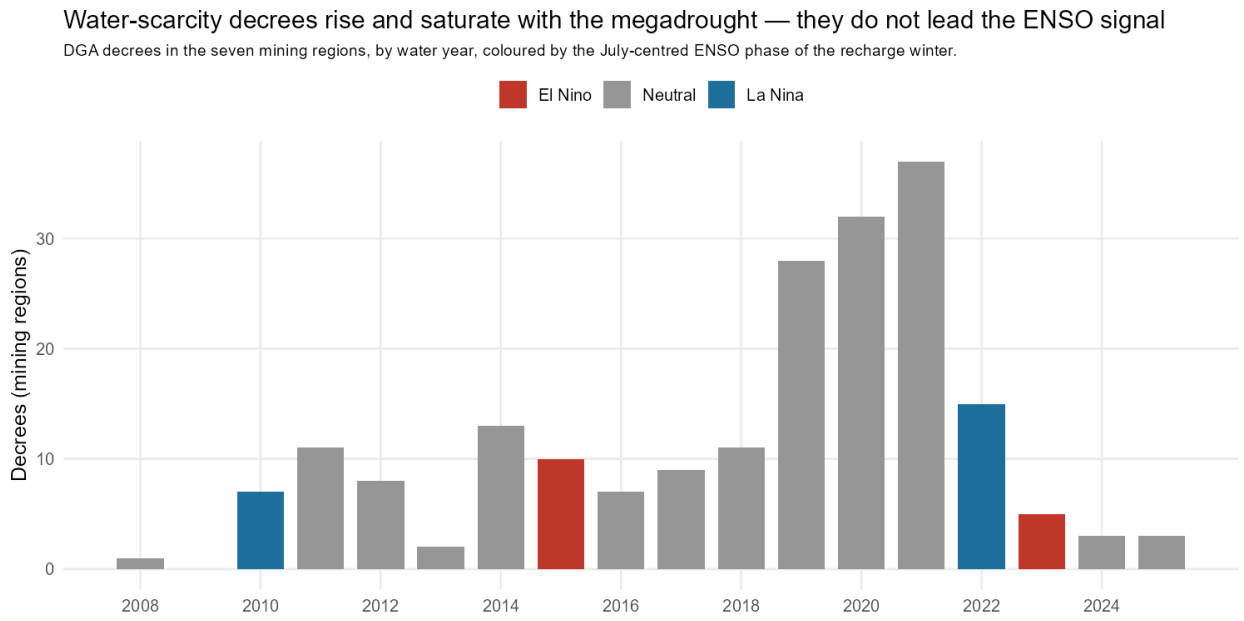


Figure 4. DGA water-scarcity decrees in the seven mining regions, by water year, coloured by the July-centred ENSO phase of the recharge winter. Decree activity climbs with the 2014–2021 megadrought and saturates — lagging the multi-year deficit rather than leading it; the El Niño years (2015, 2023) are among the quietest. Source: DGA decree registry; NOAA CPC ONI.

The response is also concentrated where the chapter's flood-exposed mines sit. The central regions (Coquimbo, Valparaíso, O'Higgins, Metropolitana) account for 189 of the 202 mining-region decrees over 2008–25; the arid northern heartland of the drought-exposed mines contributes just 13, because those operations sit largely outside the surface-water decree regime and have pivoted to groundwater and seawater (Section 4). Within the central regions, La Niña years carry about 1.75× the decree intensity of El Niño years — 2.6 versus 1.5 decrees per region-year, the expected sign — though neutral years are the most decreed of all, because most fall inside the 2014–2021 megadrought, a multi-year deficit that no single winter's ENSO state captures.

Quantity	Value
JJA ONI → physical drought stress	r ≈ −0.46 (p ≈ 0.001)
Physical drought stress → decrees	r ≈ +0.45 (p ≈ 0.001)
Direct JJA ONI → decrees (detrended)	r ≈ −0.22 (n.s.)
Central (flood) regions — decrees 2008–25	189
Northern (drought) regions — decrees 2008–25	13
Central decrees per region-year (El Niño / Neutral / La Niña)	1.50 / 2.79 / 2.62

Table 5. The ENSO → physical-drought → decree chain, the central-versus-north split, and the La Niña/El Niño decree intensity. Correlations on the mining region-year panel; see caveats. Source: companion reproduction package decree_enso_reproduction/.

The timing of the decrees confirms the same reactive posture at the within-year scale. If the regulator were acting on the forecast, central decrees would cluster in the September–November window — ahead of the dry season — just as an anticipating mine's output would (Section 2). They do not: 47% of central decrees are issued in the December–May dry season the decree itself regulates, against 27% in the Sep–Nov forecast window and 26% in winter (Figure 5). The decrees track scarcity as it manifests and renew through the multi-year megadrought (Figure 4); they are not issued ahead of it on the strength of a forecast.

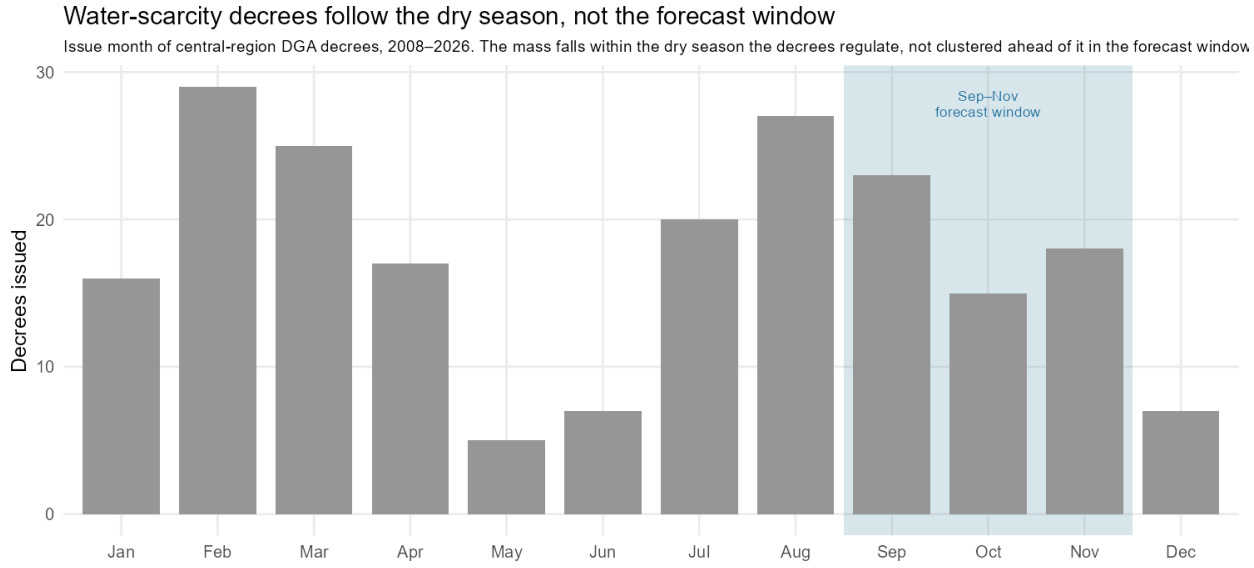


Figure 5. Issue month of central-region water-scarcity decrees, 2008–2026. The mass falls within the austral-summer dry season the decrees regulate (47% December–May), not clustered ahead of it in the September–November forecast window (27%, shaded) — the regulator reacts to realised scarcity rather than the seasonal forecast.

The regulatory record therefore tells the production and permitting story one level up. Output does not ramp into the forecast window (Section 2); permits are not filed in it (Section 3); and the State's own scarcity decrees follow the dry season and the multi-year deficit rather than anticipating them. The water-stress signal is genuinely ENSO-linked — the chain holds — but the institutional response to it is reactive and saturating, not anticipatory. Across firm and regulator alike, ENSO's months-ahead predictability leaves no anticipatory footprint; adaptation, where it happens, is structural and slow.

Caveats. The decree panel is short (eighteen water years) and the region-years are strongly serially correlated through the 2014–2021 megadrought, so the chain correlations are descriptive corroboration, not precisely estimated causal parameters — the reported p-values overstate the effective information. The megadrought is driven as much by the PDO/SAM as by ENSO, and the decree instrument is discretionary and administratively non-stationary over the decade. Full data and reproduction: decree_enso_reproduction/.

6. Conclusion

The impact season carried a majority-probability forecast signal in every episode, two to six months ahead; mines showed no observable anticipatory response to it. The poor forecastability of the precise onset — for marginal or fast-onset events — is a property of the index, not an excuse for non-adaptation: the season that actually disrupts operations was foreseeable throughout. In either case the mapping from the ENSO index to mine output is too weak, and the adaptation tools too slow, for a seasonal forecast to drive firm behaviour. What is observable is reactive and structural — contingencies activated once disruption is under way, and decade-scale water-source substitution — not anticipatory, and the pattern extends beyond the firm to the regulator, whose water-scarcity decrees follow the deficit rather than the forecast (Section 5). The test sees only what reaches monthly output, the public permitting record, and the decree registry; finer operational preparations (procurement, maintenance scheduling, water contracting, contingency planning) need not surface there, so the claim is the bounded one — no observable anticipatory response in output timing, formal permitting and capital filings, or the regulatory record. For the question this chapter asks, whether it matters that the shock is ENSO-related, the answer on the adaptation margin is that ENSO-relatedness confers no observable anticipatory advantage in practice: the predictability is real, but — on the channels we can see — unused.

Appendix A — Specification, classification, and inference

- **Output test (Section 2).** For each episode, the pre-window-to-window step — mean output deviation in the forecast-window months minus the mean in the months just before — with deviations from the neutral-ONI-year mean for the same mine and calendar month, so mine level, trend, and ordinary seasonality net out. No regression is fitted: with ~4–9 mines and 2–5 episodes a descriptive step against the neutral-year band is more honest than asymptotic inference.

- **Benchmark.** Each episode step is judged against the distribution of the same step in neutral years — the ±1 s.d. band (flood 4.5, drought 5.4); a forecast response would clear roughly ±5%.
- **ENSO-year classification.** By the decision-relevant impact season: El Niño if JJA ONI ≥ +0.5 (2015, 2023); La Niña if OND ONI ≤ −0.5 (established entering the austral-summer dry season, so the Sep–Nov window precedes the classifier). Section 1 instead keys on the onset crossing; the two definitions are used by design (Section 1).
- **Reproducibility.** Tables and figures regenerate from the companion scripts against the master ledger; ENSO datings verified against NOAA oni.ascii.txt; Section 1 probabilities from the CPC/IRI archive (extract_iri_cpc_forecasts.py); Section 5 decrees from the DGA registry (decree_enso_reproduction/).

Appendix B — Data sources and caveats

- **Sources.** Cochilco monthly ledger (output + ONI); NOAA CPC ONI (ERSSTv5); CPC/IRI consensus ENSO forecast archive; Chilean e-SEIA (seia.sea.gob.cl); COCHILCO water-consumption panel; DGA water-scarcity decree registry.
- **Coverage.** Filings are assigned by región × titular, not by filename (the raw download's filenames are unreliable — e.g. the "Andina" file actually holds Collahuasi rows). El Tesoro and Esperanza collapse to one Minera Centinela; the Codelco Antofagasta divisions are split by comuna (Chuqui/RT/MH vs Gaby) — giving 18 distinct operations, of which 16 are covered (all nine climate-exposed mines) and two excluded: Candelaria and Cerro Colorado, neither among the nine exposed mines used in the forecast-response tests, a sampling choice and not a claim about exposure (Cerro Colorado was in fact forced into closure by structural water scarcity in 2023).
- **Pre-2013 forecasts.** 2007 and 2010 probabilities come from the old IRI probabilistic table (figure 3), normalised to season-start year; modern years use the official CPC/IRI early-month consensus table.
- **Regulatory decrees (Section 5).** Short panel (eighteen water years), region-years serially correlated through the 2014–2021 megadrought, which is itself driven partly by the PDO/SAM; the decree chain is read as descriptive corroboration, not a causal estimate.
- **To add.** A full citation for the 50% threshold ("Todd 2024") and for the spring-predictability-barrier skill figures.